

To Show or Hide - Investigating the Effect of Cursor Visibility in Gaze-controlled Interfaces

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ABSTRACT

In a gaze-controlled interface, the cursor is mapped to gaze points, making the cursor visible and tracked throughout the screen. We propose a novel system where we disable the visibility of the cursor movement during the pointing task and enable selection using a separate mechanical switch. As the gaze-tracker already does the pointing task, we were not required to visualize the cursor, which might distract the user from pointing at the intended target due to offsets by the eye-gaze tracker. In this study, we compared the performance of the without-cursor tracking interface and the with-cursor tracking interface. We found that the accuracy measures, average selection time, and system usability were significantly better for the without-cursor tracking interface than the with-cursor tracking interface. We are planning to extend the study to elderly users and people with motor impairments, tics, and tremors for the performance of the without-cursor tracking interface.

CCS CONCEPTS

• **Hardware** → **Emerging interfaces**; • **Human-centered computing** → **Pointing devices**.

KEYWORDS

Eye tracker, Cursor tracking, Without cursor tracking, Mechanical Switch

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1 INTRODUCTION

The eye-tracker was evaluated as an input device for computers in the works of [8]. It was reported that the eye-tracker is simpler to use than the mouse as an input device. The eye-tracker can be used when the speed of selection is important, the size of objects selected is moderate, and when the hands are reserved for other activities. There are two tasks involved in interacting with a graphical user interface (GUI), viz., pointing and selection tasks. The pointing task is performed to inform the computer where the user wants to click. The selection task is then performed on the region they pointed at.

When we use the mouse as the human interaction device (HID), the mouse is held by the user's palm and slides over a random surface, and the movement of the mouse is mapped to the movement of the cursor, which is within the defined boundary of the display resolution. So, for the user to know where the mouse is and how much movement the mouse has made with respect to the hand movement, we visualize the mouse cursor and its movement on the display screen. This is how the pointing task works in the mouse. In the case of a gaze-controlled interface, the computer always knows where the user is looking or pointing at. So, the pointing task is already done. Now, we wanted to understand if it is required to visualize the position of the gaze on the screen using a cursor. Most [1–3, 6, 10] of the gaze-controlled interfaces map the gaze data to the mouse cursor such that the cursor is always visible and tracked based on the gaze movement by the user.

We wanted to study the performance of interaction if we did not visualize cursor movement while pointing with a gaze. As the accuracy of the eye-gaze trackers at present is not 100%, we often notice some offsets in the cursor position from our intended position on the screen where we gaze. This offset triggers users to correct it by forcefully moving the eyeballs, keeping the peripheral vision at the desired point such that the cursor moves to the desired point. This activity by users degrades the performance of interaction in addition to the strain on their eyes. As we have increased the sizes of targets on the screen enough to accommodate the small offsets by the eye tracker, the users do not have to strain to bring the cursor to the desired point. Hence, we hypothesized that by disabling the cursor tracking on the screen and just gazing at the screen and performing the selection tasks, the performance would improve by reducing the average selection time of the interaction and increasing

the accuracy measures. To support this claim, we designed two gaze-controlled interfaces, one with cursor tracking and another without cursor tracking. The following section demonstrates how the systems work. In section 3, the user study is described, followed by a discussion and conclusion in sections 4 and 5, respectively.

2 WORKING OF THE SYSTEM

We developed two algorithms to use the eye gaze-controlled interaction using with-cursor tracking and without-cursor tracking. The eye-gaze tracker was used for the pointing task, and the selection task was done using a mechanical switch. The design of the eye-gaze tracker algorithm and mechanical switch is explained in the following sections.

2.1 Working of the proposed gaze-controlled interfaces

We used an eye tracker and streamed the raw eye gaze data using the API provided. The streamed data is then filtered using the Kalman filter [9] for smoothness and scaled to match the resolution of the display. We implemented two systems, namely, gaze-controlled interface with cursor tracking and without cursor tracking. The former uses the filtered gaze data and maps the cursor position in real-time so that the cursor is always visible and tracked throughout the screen based on the user's gaze. The latter does not map the gaze data to the cursor. So, the cursor is not tracked throughout the screen. We have been used to performing pointing tasks using mouse movement and selection tasks using a switch on the mouse (mouse click). As the eye tracker does only the pointing task and there is no means to perform a selection task, we used a separate mechanical switch to perform the selection task for our gaze-controlled interfaces. Initially, we tried using the left mouse button for the selection task. However, we were not able to access it for the without-cursor tracking interface as the *set_cursor* event (for cursor position on screen) and *left_mouse_button_click* event are done at the same time in this method. This required a separate trigger switch to enable these two events together as the left-mouse button was mapped directly to only *left_mouse_button_click* event. When the user performs a selection task, the mechanical switch is connected to a micro-controller and sends the trigger data via a serial port. The algorithm of pointing and selection tasks of with-cursor tracking and without-cursor tracking is illustrated in Figure 1.

2.2 Mechanical switch design

A limit switch was connected to the Seeed Studio Seeeduino XIAOTM, which was connected to the computer via a Universal Serial Bus (USB). The case for the mechanical switch and the micro-controller was designed using FreeCADTM software and 3D printing using the Creality Ender 5 Plus 3D Printer. The CAD model is shown in Figure 2 (a), and the fabricated casing is shown in Figure 2 (b).

2.3 Experiment set-up

The participants were made to sit in a chair that had a fixed base to prevent calibration errors. The eye-gaze tracker was mounted below the monitor. The mechanical switch was placed in a comfortable position on the table for the participants to press. The screen

projected the targets in white boxes that required participants to look at them with their eyes. The set-up is shown in Figure 3.

3 USER STUDY

We conducted a user study to understand if there is any significant difference in the accuracy measures, average selection time, usability, and perceived workload for gaze-controlled interfaces using with-cursor and without-cursor tracking. The System Usability Scale (SUS) was used for subjective user experience assessment. Also, the NASA Task Load Index (NASA-TLX) was calculated to assess the perceived workload by the individuals who took part in the study.

3.1 Participants

A set of 25 participants from in and around the campus with an average age of 27.4 years (20 male and 5 female) participated in the experiment. 12 out of the total participants were wearing spectacles to correct their vision. All the necessary consents and approvals were taken.

3.2 Material

The experiments were performed on a Windows 11 PC with a 3.40 GHz Intel Core-i7 processor and 32.0 GB RAM. The display screen was 24" with a resolution of 2560 x 1440 pixels. We used Tobii Eye Tracker 5 to stream gaze data to the computer. We used a mechanical switch to perform the selection operation. The mechanical switch was connected to the computer through Seeed Studio Seeeduino XIAO via serial port. We used a set of three target sizes and three distances from the center of the screen, as listed below.

- Width of the targets: 90, 110, 120 pixels
- Distances from the center: 180, 200, 210 pixels

3.3 Design

The setup consists of an eye tracker, which tracks the eyes of the participant and maps them on the computer screen. A standard procedure for the target positioning for the gaze-controlled interface with cursor tracking and without cursor tracking was executed by the participants. The selection was made using the mechanical switch. The pointing task that the participants were asked to perform was similar to the ISO 9241-9 pointing task [5, 7]. It had the target object in white colour and the obstacles or the distractors in blue colour. This is shown in Figure 4. It was ensured that the cursor was brought to the centre of the screen before each target was clicked. This was done to note the distance and time that the participants took to select the target. In the first experiment, the participants were asked to follow the cursor with their eyes and click on the targets using the mechanical switch. In the second experiment, the participants were asked to see the targets (the cursor was not visible) and click the mechanical switch. These trials were conducted in alternating sequences for the succeeding participants to reduce the order effect.

3.4 Procedure

The pointing task was performed using the eye tracker in two modes: with and without a cursor on the screen. The participants

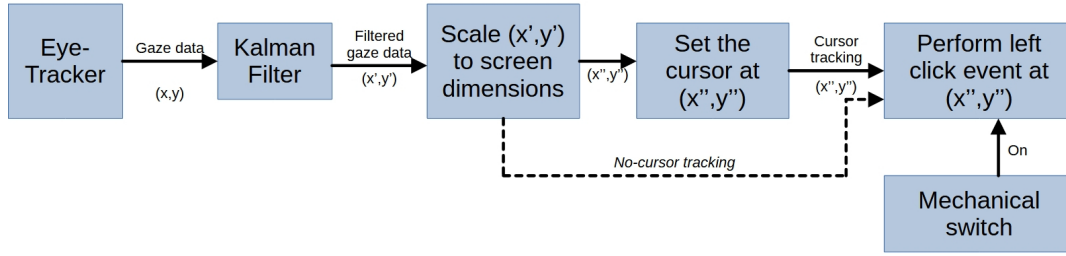


Figure 1: The working of the with-cursor and without-cursor tracking algorithms

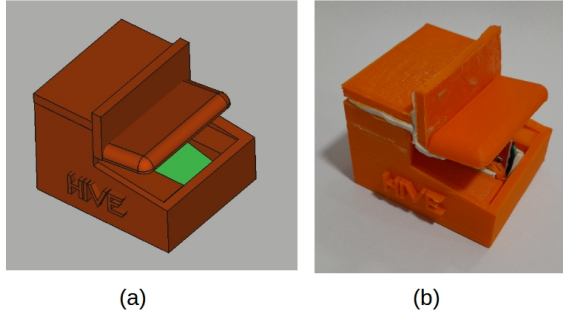


Figure 2: (a) CAD model of the mechanical switch (b) The switch in the 3D printed casing

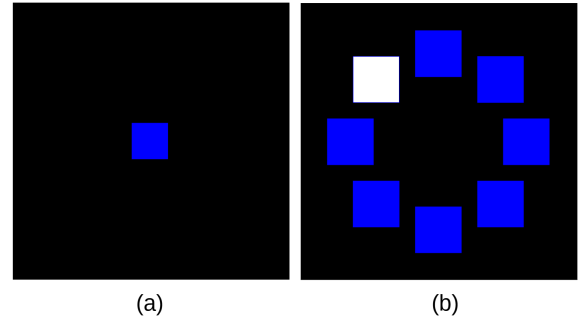


Figure 4: The screenshot of ISO 9241-9 pointing task (a) the blue square at the center, which is treated as a reference point (center selection), and (b) the white square is the target that the user must click (target selection)

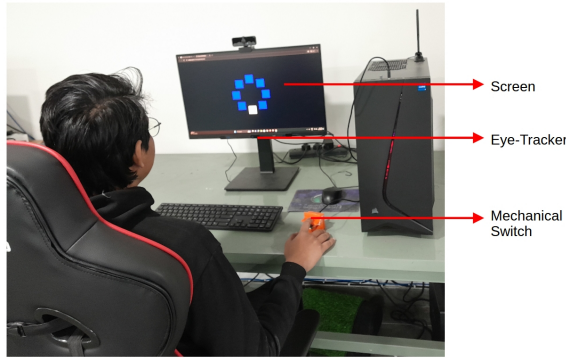


Figure 3: The experiment setup where the screen, eye-tracker, and mechanical switch are shown

were asked to complete the consent form before participating in the experiment. Tobii's calibration software calibrated the eye tracker for each participant. Then, the participants were briefed about the experiment. The participants were given free trials to familiarize themselves with the interface. In two trials, they were asked to perform ISO pointing tasks using with-cursor tracking and without-cursor tracking interfaces. After each trial, they were asked to complete the System Usability Scale (SUS) and NASA Task Load Index (NASA-TLX) assessments. Each trial took less than 5 minutes, with 45 target hits per trial per participant.

3.5 Results

In this section, we discuss the results in terms of accuracy measures, average selection time, system usability scale (SUS) assessment, and NASA Task Load Index (NASA-TLX) assessment. They are as follows:

3.5.1 Accuracy measures for the experiment with and without cursor tracking. In the measurement of accuracy (as done in [5]), the pointer path is characterized by discrete events and continuous measures. The discrete events include target re-entry, task axis crossing, movement direction change, and orthogonal direction change. The continuous measures include movement variability, movement error, and movement offset. Table 1 shows the means and standard deviations for the with-cursor and without-cursor tracking trials and their significance values.

- (1) **Target re-entry (TRE):** The target re-entry occurs when the pointer enters, leaves, and re-enters the target region. In our work, the TRE is present in with-cursor experiments but completely absent in without-cursor experiments.
- (2) **Task axis crossing (TAC):** The task axis crossing occurs when the pointer crosses the task axis to reach the target region. The TAC is present in with-cursor experiments but absent in without-cursor experiments.
- (3) **Movement direction change (MDC):** The direction of the pointer with respect to the task axis changes three times before reaching the target. This difference in these data was not

normally distributed as per the Shapiro-Wilk method. So, the Wilcoxon Signed Rank test was done where the significance was $p < 0.01$. Thus, the MDC for without-cursor tracking is significantly lower than that of with-cursor tracking trials.

- (4) **Orthogonal direction change (ODC):** The direction of the pointer changes direction twice about the axis orthogonal to the task axis. The difference in these data was not normally distributed, and the Wilcoxon Signed Rank test showed that significance was $p < 0.01$. Therefore, the ODC for the without-cursor experiment is significantly lower than the with-cursor experiment.
- (5) **Movement variability (MV):** The difference in the distances between the sample points that lie in the straight line about an axis that is parallel to the task axis is known as MV. The difference in data is not normally distributed, and the significance was $p < 0.01$. Therefore, the MV for the without-cursor experiment is significantly lower than the with-cursor experiment.
- (6) **Movement error (ME):** This is the recording of the average deviation of sample points with respect to the task axis, where the points can be below or above the axis. The statistical results were the same as in the previous case. The without-cursor experiment had a significantly lower score than the with-cursor experiment.
- (7) **Movement offset (MO):** Considering the task axis, the average deviation of the sample points from them is known as MO. Their statistical results remained the same as in the previous case. Therefore, for MO, the without-cursor experiment, the MO is significantly lower than the with-cursor experiments.

3.5.2 Average selection time. The average selection time was calculated for each participant for each trial. It was done by taking an average selection time (in milliseconds) for the 45 iterations (with and without-cursor tracking) when the study was conducted. It was observed that it took an average time of 2303 ms for the participants to select the target using the with-cursor-tracking interface. However, it took 1404 ms for the participants to use the without-cursor tracking interface. The differences in the average selection time (in two study modes) were calculated and tested for normality using the IBM SPSS. The Shapiro-Wilk method showed a significance of $p < 0.05$, showing that the data is not normally distributed. Therefore, the non-parametric test is done using the Wilcoxon Signed Rank test. The results indicate that the average selection time of 1404 ms in the without-cursor-tracking interface is significantly less than that of the cursor-tracking interface. The index of difficulty (ID) vs. average selection time plot is shown in Figure 5 [4].

3.5.3 System Usability Scale (SUS) Assessment. The average SUS score was calculated for both trials of the experiment, considering all the participants. The with-cursor trial had an average SUS score of 74.1, and the without-cursor had a score of 83. A test of normality was performed in which the Shapiro-Wilk method showed a significance $p < 0.05$, indicating that the data is not normally distributed. Hence, the Wilcoxon Signed Rank test was performed, and the results indicated significant differences in the SUS score. The without-cursor experiment performed significantly better.

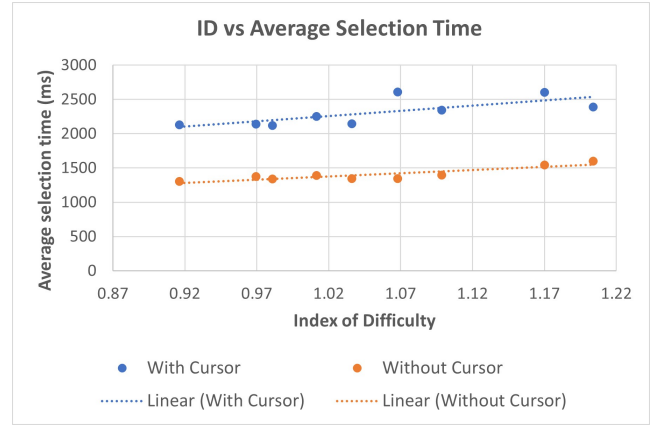


Figure 5: Index of difficulty (ID) vs. average selection time

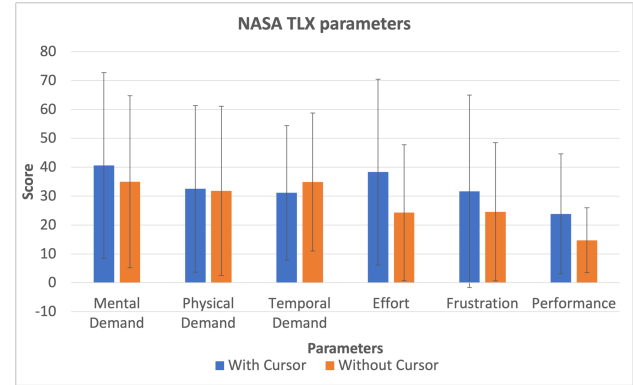


Figure 6: NASA TLX parameters for with-cursor and without-cursor tracking

3.5.4 NASA Task Load Index (NASA-TLX) assessment. The average TLX score, considering all the participants in the two experiments, was calculated. The average TLX score was 33.0 for with-cursor tracking experiments and 27.5 for without-cursor tracking experiments. The differences in TLX scores recorded for individual participants were normally distributed. Therefore, the paired T-test was performed, indicating no significant difference in the TLX score of both experiments. Also, the individual parameters of the TLX scores, such as mental demand, physical demand, temporal demand, effort, frustration, and performance, were checked for significant differences in scores in both experiments. It was found that there is no significant difference in the score of any of these individual parameters. Their mean and standard deviation are shown in Figure 6.

4 DISCUSSION

From the accuracy measures, we can interpret that the participants performed the without-cursor tracking interface with improved accuracy compared to the with-cursor tracking interface. We can also infer from the results that the TRE and TAC are zero for without-cursor tracking interfaces. This is because the cursor was never

Table 1: The means and standard deviations of the accuracy measures

Variable	With-Cursor		Without-cursor		Significance
	Average	Standard Deviation	Average	Standard Deviation	
Target re-entry (TRE)	0.0613	0.1237	0	0	absent in without-cursor experiment
Task axis crossing (TAC)	1.1153	0.2660	0	0	absent in without-cursor experiment
Movement direction change (MDC)	1.3274	1.1864	0.4474	0.1927	Wilcoxon Signed Rank Test ($p < 0.01$)
Orthogonal direction change (ODC)	3.6939	1.1173	0.4419	0.1871	Wilcoxon Signed Rank Test ($p < 0.01$)
Movement variability (MV)	4.0026	5.8921	0.0001	2.36743E-05	Wilcoxon Signed Rank Test ($p < 0.01$)
Movement error (ME)	5.2874	7.3784	0.0002	3.069E-05	Wilcoxon Signed Rank Test ($p < 0.01$)
Movement offset (MO)	6.4299	8.6800	0.0002	3.81041E-05	Wilcoxon Signed Rank Test ($p < 0.01$)

moved when the participant kept hovering at the targets on the screen. The results show that the average selection time for the without-cursor tracking interface was significantly less than that for the with-cursor tracking interface. This shows that the participants quickly performed the selection task in the without-cursor tracking interface. From the ID vs. average selection time plot, it is found that for every difficulty level, the without-cursor tracking interface took the least time compared to the with-cursor tracking interface. The SUS score indicates that the without-cursor tracking interface had better usability than the with-cursor tracking interface. Though we found no significant difference in the TLX score, the parameters, namely mental demand, physical demand, effort, and frustration, were relatively lesser for the without-cursor tracking interface. Although the participants were English-speaking, English is not the native language of any of the participants. This could have led the participants to misinterpret the inherent meaning behind the subjective questions asked in the NASA TLX questionnaire, which might have degraded the TLX scores. We also found that the range in which each participant gave scores in TLX had a huge variance among different participants.

We encountered a situation where the eye tracker did not successfully pass through the calibration for a particular participant. This might be because the participant had squinting eyes. We also found certain participants gazing at the targets for longer durations at the beginning of the trail in a without-cursor tracking interface. This might be due to two reasons. They might have thought that the experiment was with a cursor tracking interface, and so might have waited for the cursor to appear at the target. It might also have been the case where they might have been not sure if they gazed at the right target as we did not give any visual feedback for the gaze position on the screen. However, they started performing better over time while undergoing the trial. This might be the effect of the participants' learning curve for this new interface. Despite the discrepancies in the TLX score, we found significant improvement in the overall performance of the without-cursor tracking interface.

As our without-cursor tracking interface does not require people to use their hands for pointing tasks, which require certain precision

and accuracy in hand movement, we are planning to extend the study to elderly users, people with motor impairments, tics, and tremors. We also plan to build a new interface like "windows, icons, menus, pointer" (WIMP) for optimized gaze-controlled interaction without cursor tracking.

5 CONCLUSION

In this work, we performed a comparative study using an eye-tracker and a mechanical switch for with-cursor and without-cursor tracking interfaces. We conducted a user study with 25 participants and evaluated the two types of gaze-controlled interfaces. The performance metrics like the accuracy measures, average selection time, SUS score, and NASA TLX score were analyzed. The accuracy measures, average selection time, and the SUS score performed significantly better for the without-cursor tracking interface. The NASA TLX score did not show any significant difference between the two interfaces. From the overall results, we infer that the without-cursor tracking interface has a significant potential to be used in any gaze-controlled interface devices with improved performance. We are extending the study to analyze the subjective TLX scores for discrepancies. We also plan to use our proposed without-cursor tracking interface to repeat the study with different eye trackers for their performances.

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